10.3 Using Chords

Geometry 10.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- use congruent chords to find arc measures.
- use a diameter that is perpendicular to a chord to find lengths.
- use the fact that congruent chords are the same distance from the center.

Vocabulary · fill in the missing word

Chord — a segment whose _____ are on a circle

Perpendicular bisector — a line that is perpendicular to a segment at its _____

Equidistant — the same _____ from

Distance from a point to a line — the length of the _____ segment from the point to the line

Part 1 · Discovery: find the center by folding

Draw a chord on your paper circle. Fold so its endpoints match and crease. Repeat with a second chord. Mark where the creases cross.

Distance from the crossing point to the circle: 1st place	2nd place	3rd place

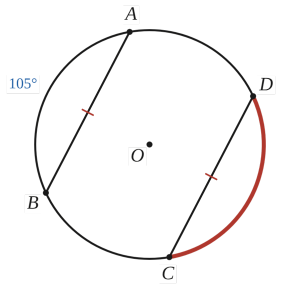
The crease of a chord always passes through the _____ of the circle.

Part 2 · Notes and examples (fill in each note, then work the example)

Congruent Corresponding Chords Theorem: in the same circle or congruent circles, two minor arcs are congruent if and only if their corresponding _____ are congruent.

Example 1

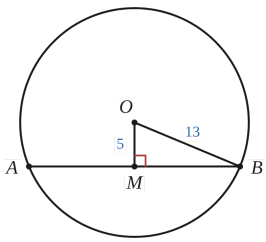
In circle O, $\overline{AB} \cong \overline{CD}$ and m(arc AB) = 105°. Find m(arc CD).



Perpendicular Chord Bisector Theorem: if a diameter of a circle is perpendicular to a chord, then the diameter _____ the chord and its arc.

Example 2

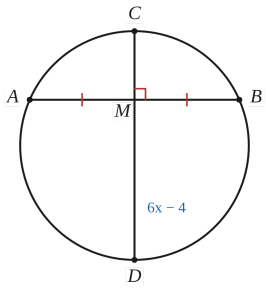
Circle O has a radius of 13. Chord $\overline{AB}$ is 5 units from the center. Find AB.



Perpendicular Chord Bisector Converse: if one chord is a perpendicular bisector of another chord, then the first chord is a _____.

Example 3

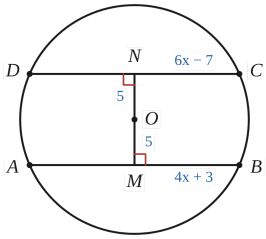
Chord $\overline{CD}$ is the perpendicular bisector of chord $\overline{AB}$. The radius of the circle is 13, and CD = 6x − 4. Find x.



Equidistant Chords Theorem: in the same circle or congruent circles, two chords are congruent if and only if they are _____ from the center.

Example 4

Chords $\overline{AB}$ and $\overline{CD}$ are each 5 units from the center of the circle. AB = 4x + 3 and CD = 6x − 7. Find CD.



Example 5

Only the edge of a round garden is left. Explain how to find its center using chords.

Part 3 · Partner practice

Partner 1

A circle has a radius of 10. A chord is 12 units long. How far is the chord from the center?

Partner 2

Two chords are the same distance from the center of a circle. Their lengths are 7x − 1 and 5x + 9. Find x and the length of each chord.